

## PROJECT STATUS / COLLABORATION BRIEF

# MEHLISSA Next

A reproducible multilayer research platform for body transport, molecular interaction, cell response, and Nano-IoT communication

| PLATFORM                   | ACCEPTED DELIVERY  | CURRENT FOCUS                                                                         |
|----------------------------|--------------------|---------------------------------------------------------------------------------------|
| Integrated research system | M0-7 and UX-1-UX-6 | PCQ-1 pulmonary and capillary pre-outcome amendment and governed evidence acquisition |

Prepared for prospective contributors and research partners

Status date: 5 September 2026

Branch: mehlissa-next-generation | Workbench 1.0.0 | Published CI verified

Repository: [github.com/RegineWendt/MEHLISSA](https://github.com/RegineWendt/MEHLISSA)

## Executive summary

MEHLISSA Next has progressed from a set of valuable historical prototypes to a coherent, modular research simulation platform. Its scientific simulation stack, research-use interfaces, and assurance mechanisms together constitute the platform. The scientific stack now implements the complete architectural path envisioned for the first dissertation-driven demonstrator: systemic transport, a replaceable lung model, capillary transport and molecular channels, receptor and intracellular cell response, nanodevice communication, an active gateway, a body area network, and external analysis.

**Key achievement:** All major layers can now participate in one reproducible FP9/lung fingerprinting workflow while remaining independently replaceable behind explicit contracts.

- Milestone gates M0 through M7 have passed their documented acceptance reviews.
- The implementation uses C++20, CMake, vcpkg, strict JSON Schemas, typed SI quantities, stable error contracts, deterministic random streams, and explicit provenance.
- The published Workbench 1.0 commit passes all 286 local Windows/MSVC tests and GitHub Windows/MSVC, Linux/GCC, and Linux/Clang CI. The Clang path also passes formatting, clang-tidy, AddressSanitizer, and UndefinedBehaviorSanitizer.
- The native macOS/Apple Clang CMake path, automatic Workbench executable discovery, and required macOS-15 ARM64 GitHub job are accepted by GitHub CI run 33956456353. The path produces a source-built command-line simulator, not a .app bundle.
- The integrated research-use layer provides a validated five-family, ten-example catalog; safe scenario editing and validation; direct and campaign execution; non-overwriting reports; process-based Python readers; optional plots; and two notebooks.
- MEHLISSA Research Workbench 1.0 provides graphical discovery, guided configuration, corrective validation, guarded execution, result comparison, provenance and evidence audit, descriptive campaign analysis, exact-value export, isolated installation, and tested accessibility and recovery paths. It calls the accepted C++ and Python interfaces and does not create a second scientific implementation.
- English user, architecture, model, traceability, and decision documentation supports international collaboration.
- A reviewable Paper 1 platform/methods candidate now binds a locked technical protocol, a schema-validated six-family evidence matrix, three complete experiment archives, source export, claim registry, and SHA-256 inventory.
- The next scientific cycle has started with PCQ-1 design v0.1.0. It freezes the entering lung and capillary artifacts by SHA-256, a bounded non-clinical claim, four separately decidable tracks, six primary endpoints, no-refit and source-disjointness rules, six uncertainty classes, seven negative controls, and an amendment gate before new participant-level outcomes are inspected. PCQ-1.2 now adds a machine-checked thirteen-candidate source register with track rankings, access and rights boundaries, measurement jointness, source overlap, rejected-primary reasons, and transparent public-outcome exposure. No request has been sent and no candidate participant record acquired. This is prospective study discipline and governed source selection, not yet new physiological evidence.
- The result is a reproducible research-software demonstrator. It is not a clinical assay model, a medical device, or a patient-specific digital twin.

| Dimension              | Assessment                                                                                                          |
|------------------------|---------------------------------------------------------------------------------------------------------------------|
| Software architecture  | Strong modular foundation with explicit ownership, conservation, lifecycle, configuration, and evidence boundaries. |
| End-to-end integration | Complete first software vertical slice through fingerprinting Levels A-E.                                           |

| Dimension                            | Assessment                                                                                                                                                                                                                                         |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scientific validation</b>         | Mixed maturity: verified equations, literature-parameterized candidates, selected independent comparisons, and multiple synthetic mechanisms.                                                                                                      |
| <b>Integrated research-use layer</b> | Commands, discovery, reports, campaigns, Python and notebook access, and Workbench 1.0 form the supported delivery surface of MEHLISSA. They expose the scientific stack without duplicating its schemas, models, validation, or result authority. |
| <b>Clinical readiness</b>            | Not claimed. Patient prediction, diagnosis, and treatment recommendations are explicitly outside the present scope.                                                                                                                                |

## Requirements in two dimensions

The 83 baseline requirements now carry separate implementation and evidence statuses. This prevents software completion from being read as scientific validation. Functionally, 58 requirements are complete, 15 are partly implemented, five remain legacy-only, and five are specified but not yet implemented. For evidence, 39 are verified for their declared bounded claim, 34 have partial evidence, eight are unverified in MEHLISSA Next, and two need a dedicated research evidence programme.

DONE/PART is an important and legitimate state: executable behavior and its software contracts are present, but physiological, biological, external-data, scale, or release-wide qualification remains incomplete. VERIFIED refers to the verification mode required by the individual requirement; it does not automatically mean clinical or participant-level validation. The canonical row-level assessment and remaining gaps are maintained in the Traceability Matrix.

## How to read project labels

Short identifiers preserve traceability across requirements, code, tests, and evidence. They are not assumed knowledge in this report.

| Label                    | Plain-language meaning                                                                                                                                                                                                                                                                                                                    |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>M0 through M8</b>     | Sequential milestone gates. M7 is the accepted complete fingerprinting software demonstrator. The future M8 gate concerns a governed research digital twin.                                                                                                                                                                               |
| <b>M7.4</b>              | A numbered increment within a milestone. M7.4 adds concentration- and receptor-binding-based fingerprint detection.                                                                                                                                                                                                                       |
| <b>Level A</b>           | Historical-timing baseline for the fingerprinting scenario; it reproduces published dissertation event semantics and is not a separate "A run".                                                                                                                                                                                           |
| <b>Level B</b>           | Mechanistic fingerprint detection based on concentration, receptor binding, and a threshold.                                                                                                                                                                                                                                              |
| <b>Level C</b>           | Explicit release and complete/incomplete assembly of nine fingerprint tiles.                                                                                                                                                                                                                                                              |
| <b>Level D</b>           | Executed local, gateway, body-area-network, and external-station communication.                                                                                                                                                                                                                                                           |
| <b>Level E</b>           | Sensitivity, specificity, false-positive/false-negative, robustness, and misclassification analysis.                                                                                                                                                                                                                                      |
| <b>P0 through P3</b>     | Roadmap priority tiers: indispensable foundation, dissertation core, research-platform expansion, and long-term vision. They are not completion states.                                                                                                                                                                                   |
| <b>DONE/PART</b>         | Two-dimensional requirement status: functional implementation is complete, while the required evidence is only partial. Other rows independently combine DONE, PART, LEGACY, or SPEC implementation with VERIFIED, PART, UNVERIFIED, or RESEARCH evidence.                                                                                |
| <b>UX-1 through UX-6</b> | Delivery packages used to build MEHLISSA's cross-cutting research-use layer, from one-command execution to a graphical research workbench. They were implemented after M7, but their accepted outputs are integral platform capabilities rather than a separate add-on. UX-6.1 through UX-6.8 are reviewable increments of the workbench. |

| Label                      | Plain-language meaning                                                                                                                                                                                             |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>P1-E1 through P1-E3</b> | Predeclared Paper 1 technical experiments: body-observation neutrality/resources, small M7 resource/replay behavior, and CLI/Python/Workbench result parity. They are experiment identifiers, not evidence grades. |
| <b>run</b>                 | One reproducible execution of an experiment or scenario. A qualified term such as “baseline run” must state what makes that execution distinct.                                                                    |

FP9 is the nine-part molecular fingerprint used by the first complete scenario. BVS is the historical BloodVoyagerS whole-body distribution reference. BAN is the body area network between gateway and external station. ODE denotes an ordinary differential-equation model, and SSA denotes a stochastic simulation algorithm.

## 1. Conceptual system architecture

MEHLISSA separates the research question from the models used to answer it. A scenario selects versioned model cards, initializes independent components, coordinates explicit transfers and events, and produces bounded results with provenance. This permits a fast surrogate, population model, field model, or detailed particle model to be compared without embedding medical special cases in the kernel.

The implemented conceptual path is:

1. **Body:** systemic circulation and whole-body transport.
2. **Lung or another organ:** replaceable organ-specific models.
3. **Capillary bed:** local transit, exchange, entity fate, and molecular channels.
4. **Cell:** receptor binding, intracellular signaling, delivery, and response.
5. **Nano-IoT:** nanodevice endpoints, local links, bounded relays, and gateway.
6. **BAN and station:** external measurement and a governed return command.

That scientific path is one of three integrated aspects of MEHLISSA:

1. **Scientific simulation stack:** the body, organ, capillary, cell,

Nano-IoT, BAN, and station components that represent the investigated mechanisms.

2. **Research-use layer:** the command line, catalog, reports, campaigns,

Python readers, notebooks, and Workbench that let researchers configure, execute, inspect, compare, and share experiments.

3. **Assurance layer:** schemas, typed units, provenance, evidence boundaries,

deterministic tests, gate reviews, and CI that keep both other aspects reproducible and auditable.

The historical M0-M7 gates primarily record growth of the scientific stack and its runtime contracts. UX-1 through UX-6 record delivery of the research-use layer. This distinction is useful for traceability, but neither sequence alone defines the product: MEHLISSA is the integrated platform formed by all three aspects.

Binding engineering principles are:

- **Layer independence:** body, organ, capillary, cell, and communication models live in separate libraries.
- **Replaceable resolution:** model variants share stable boundaries while documenting their own validity scopes.
- **Explicit ownership and conservation:** identities and physical amounts cannot silently disappear or be duplicated at a boundary.
- **Dimension-safe internals:** public numerical APIs use typed quantities; files state units explicitly.
- **Reproducibility:** runs bind time, seed, named random streams, model

versions, hashes, logs, and checkpoints.

- **Strict configuration:** versioned JSON Schemas and semantic loaders reject unknown or inconsistent inputs.
- **Evidence before claims:** verification, calibration, independent validation, historical regression, and sensitivity analysis remain distinct.
- **Bounded scale:** populations, aggregates, and record limits prevent uncontrolled output growth.

Independent MEHLISSA Next code is licensed under MPL-2.0. Legacy code and direct ports remain GPL-2.0-only. New original documentation and approved original data are licensed under CC BY 4.0. Artifact-specific provenance and license sidecars prevent unclear data rights from entering public packages.

## 2. Milestones achieved: M0-M3

| Gate | Plain-language result                      | Representative capabilities                                                                                                                                                                                                                                              |
|------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M0   | Project charter and architecture decisions | Dissertation requirements and traceability; new-kernel decision; four-layer architecture; lung selected as first organ; licensing, data inventory, users, workflows, validation gaps, and partner roles.                                                                 |
| M1   | Trustworthy simulation kernel              | Monotonic nanosecond time; 3D position; typed SI units; named deterministic random streams; component lifecycle and ownership; experiment manifests; provenance; structured errors and JSONL logs; checkpoints; byte-stable cross-platform reference.                    |
| M2   | Validated body transport layer             | Schema-validated 95-segment vascular graph; approved legacy migration; graph and vessel-9 branching invariants; injection, extraction, and identity-preserving transport; bounded observations; rest, exercise, and posture overlays; 6,359/63,590-agent BVS regression. |
| M3   | Body-organ coupling and lung reference     | Typed body-organ transfers; coarse, serial-region, pulmonary OD, flow-adaptive, age-conditioned, pressure-distensible, and five-lobe parallel-bed models; source-disjoint validation paths; same-scenario resolution comparison; historical FP9 timer baseline.          |

M0-M3 establish that a medical scenario no longer needs its own simulation engine or vascular implementation. The kernel remains generic, a vascular graph can change without recompilation, and a caller can select a lung model at a declared resolution while preserving the meaning and ownership of transported entities and quantities.

The pulmonary models include literature-parameterized and independently evaluated endpoints, but remain zero-dimensional or equivalent representations. They do not yet provide anatomical one-dimensional flow, pulsatile pulmonary mechanics, patient imaging, or a complete jointly measured validation cohort.

## 3. Milestones achieved: M4-M7

| Gate | Plain-language result         | Representative capabilities                                                                                                                                                                                                                                                                                          |
|------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M4   | Capillary communication layer | Executable arteriole-capillary-venule route; geometry and continuity closure; dynamic recruitment; balanced blood/endothelium/interstitium/cell exchange; residence and terminal disposition; analytical diffusion, Brownian endpoint, trajectory, radial finite-volume, and shared axial advection-reaction models. |

| Gate | Plain-language result                  | Representative capabilities                                                                                                                                                                                                                                                                                                                       |
|------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M5   | Cell response layer                    | Analytical and time-varying receptor-ligand binding; stochastic finite-receptor SSA; capillary-to-cell signal hand-off; shared intracellular ODE/SSA network; conservative device release and uptake; synthetic apoptosis event; cohort-compressed populations up to one trillion represented cells.                                              |
| M6   | End-to-end Nano-IoT plane              | Nanodevices and local messages; one-hop delivery outcomes; bounded routing and relays; active gateway uplink/downlink; BAN and governed station loop; payload-free external network-simulator adapter; twelve fail-closed resilience and boundary-misuse cases.                                                                                   |
| M7   | Holistic fingerprinting vertical slice | Strict selection of 13 scenario artifacts; ten-stage causal contract; typed runtime; identity trace; artifact SHA-256 hashes; concentration-driven binding and negative control; nine FP9 tiles and incomplete control; locator-to-station communication; Level-E classification metrics and Wilson intervals; deterministic result schema 2.0.0. |

In one sentence: MEHLISSA can now execute a reproducible software workflow in which a configured multilayer stack produces a biological detection event, assembles an explicit fingerprint, communicates it through local and external boundaries, and reports the outcome with complete artifact identity and declared uncertainty limitations.

Positive, negative, invariant, numerical, integration, regression, and schema tests cover all accepted milestones. All compiler and analysis CI jobs pass. Gate reviews explicitly prevent a software pass from being presented as physiological or clinical validation. The English User Guide is mandatory review evidence at every future milestone and UX-package review.

## 4. How the integrated MEHLISSA platform can be used today

| Access route                           | Suitable uses                                                 | Current examples                                                                                                                                                                                                   |
|----------------------------------------|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MEHLISSA Research Workbench 1.0</b> | Guided use by new and experienced researchers                 | Discover models and examples; edit, validate, and run FP9/lung scenarios; run the curated campaign; inspect results, comparisons, provenance, evidence, and descriptive analyses; export exact tables and figures. |
| <b>Command line</b>                    | Scriptable and transparent direct use without writing C++     | Validate manifests and vascular graphs; run the minimal workflow and BVS regression; apply body-state overlays; discover, validate, execute, summarize, and report the complete M7 demonstrator.                   |
| <b>Python API and notebooks</b>        | Reproducible analysis and integration into research workflows | Launch accepted processes; read versioned scenario and campaign results; create optional plots; reproduce the two licensed notebook workflows.                                                                     |
| <b>Reference and benchmark drivers</b> | Reproduce accepted scientific and numerical comparisons       | Pulmonary validations; coarse-versus-five-lobe comparison; molecular-channel comparisons; historical FP9 timing; benchmark campaigns.                                                                              |
| <b>C++ component APIs</b>              | Compose or extend advanced research workflows                 | Organ-capillary round trip; cell and communication models; external simulator adapter; direct access to the same M7 runtime used by the CLI.                                                                       |
| <b>Automated test suite</b>            | Audit contracts and reproduce gate evidence                   | Focused M1-M7 CTest filters, negative controls, deterministic replay, conservation, and identity checks.                                                                                                           |

Representative research questions include:

- How does injection site or physiological state affect systemic distribution and transit?
- Which outputs change when a coarse lung surrogate is replaced by five parallel pulmonary lobe beds?
- How do binding, intracellular signaling, device release, uptake, and a synthetic cell response connect without violating amount ownership?
- How do loss, corruption, expiry, replay, routing errors, or station-policy rejection affect a Nano-IoT measurement path?
- Can the complete FP9/lung workflow preserve causal identity, report incomplete fingerprints, and compute classification metrics reproducibly?

After installing the Workbench package and building the matching simulator, the integrated graphical route starts with:

```
mehlissa-workbench --repository-root . --check
mehlissa-workbench --repository-root .
```

The first command checks the repository, simulator, schemas, examples, and other prerequisites. The second starts the local Workbench and opens its address in the browser. The complete M7 scenario also remains available through the scriptable command-line route:

```
build/windows-msvc/apps/Debug/mehlissa.exe scenario list
build/windows-msvc/apps/Debug/mehlissa.exe scenario run --file
examples/scenarios/fp9-lung-level-a-v1.json --output results/fp9-reference
```

The separate `scenario validate` command checks the same inputs without executing them. The output argument is a parent directory; each run creates a unique UTC-labelled child containing `result.json`, `provenance.json`, `run.log.jsonl`, and `summary.txt`; the application prints the exact path. An existing result can be summarized with `mehlissa result summarize --file <result.json>`. The command reuses the accepted M7 composer and holistic runner, validates all thirteen selected definition/schema pairs before execution, and preserves the non-clinical interpretation boundary.

The integrated discovery commands use the same catalog and do not create a second source of scientific truth:

```
mehlissa model list
mehlissa model describe --id organ.pulmonary-zero-dimensional
mehlissa example list --model organ.pulmonary-zero-dimensional
mehlissa example copy --id scenario.fp9-complete --output work/fp9-study
```

The strict versioned catalog describes five model families and ten licensed starters, including maturity, validity, evidence, limitations, key parameter paths, artifacts, and documentation. Checks reject duplicate IDs, unknown model references, missing assets, and paths outside the repository. Copying preserves the license and refuses to overwrite work. The complete published platform passes all 286 local Windows/MSVC tests and all supported GitHub CI jobs.

## 5. Current personalization capability

MEHLISSA supports parameterized research profiles, but not a complete virtual person. A contributor can derive a versioned model card, state whether values are measured, literature-derived, calibrated, assumed, or used only for sensitivity analysis, and preserve units, uncertainty, provenance, and limitations.

| Scope       | Available now                                                                                  | Not yet available                                                                                     |
|-------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Body</b> | Load any schema-valid vascular graph; apply compatible cardiac-output and transition overlays. | Production import of segmented personal anatomy; complete pressure/compliance and disease physiology. |

| Scope            | Available now                                                                                                  | Not yet available                                                                                                                       |
|------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Lung</b>      | Configure pressure boundary, resistance, compliance, flow, transit, age band, distensibility, and lobe shares. | Automatic parameter inference; geometry-resolved patient lung; complete personal validation.                                            |
| <b>Capillary</b> | Configure equivalent geometry, flow, recruitment, exchange, observation, and disposition.                      | Patient microvascular anatomy, hematocrit/rheology, qualified kinetics, and spatial tissue coupling.                                    |
| <b>Cell</b>      | Configure channel, receptor, reaction, delivery, threshold, response, and cohort parameters.                   | Patient-specific biomarker calibration, mechanistic apoptosis, and disease-specific validated families.                                 |
| <b>Whole run</b> | Fix seeds, versions, checksums, output paths, and interpretation limits.                                       | Canonical patient manifest, automatic cross-layer composition, parameter estimation, uncertainty propagation, and longitudinal updates. |

Identifiable patient data must not be committed to the repository. Clinical or participant-specific work requires institutional approval plus explicit consent, pseudonymization, retention, access-control, and provenance workflows. The future M8 Research Digital Twin gate still forbids clinical decision claims without an appropriate regulatory and validation path.

## 6. Current scientific limitations

| Area                        | What is demonstrated                                                                         | What remains unproven                                                                                               |
|-----------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Cross-layer timing</b>   | One deterministic causal workflow with historical timing evidence.                           | Anatomically resolved localization and collector-return timing predicted by the coupled models.                     |
| <b>Fingerprint biology</b>  | Concentration-driven receptor detection and explicit target identity.                        | Qualified FP9 gene-product combination, disease-marker data, and biological parameter validation.                   |
| <b>Tile assembly</b>        | Nine explicit tile identities, an all-required rule, and incomplete negative control.        | Executed NetTAS or molecular self-assembly physics and independently validated assembly kinetics.                   |
| <b>Communication</b>        | Replaceable links, routing, gateway, BAN, station, metrics, and external-simulator boundary. | Calibrated molecular, intrabody, wearable, Bluetooth, or IEEE 802.15.6 behavior.                                    |
| <b>Classification</b>       | Correct sensitivity/specificity accounting, labelled cases, and Wilson intervals.            | Empirical prevalence, clinical sensitivity/specificity, patient variability, and diagnostic utility.                |
| <b>Physiology</b>           | Literature-scoped lung models and selected independent endpoint comparisons.                 | Jointly measured cohorts, broad disease states, pulsatility, anatomy, hematocrit, and full uncertainty propagation. |
| <b>Therapeutic response</b> | Conservative release/uptake and a bounded synthetic cell-response event.                     | Mechanistic pharmacology, spatial drug transport, metabolism, toxicity, and treatment efficacy.                     |

**Interpretation rule:** A verified software contract shows that the declared mechanism is implemented consistently. It does not, by itself, show that the mechanism predicts a biological or clinical outcome.

## 7. Remaining development program

The evidence-and-validity inventory and the first Paper 1 technical baseline are no longer future work. Candidate paper1-platform-methods-rc1-20260903 contains the six-family matrix, protocol v2.0.0, 112-attempt body campaign, small repeated M7 resource study, access-path parity check, complete raw archives, source export, hashes, and a claim registry. The suggested tag is paper1-platform-methods-rc1; it has not been created, and no DOI or final release is implied. The remaining programs begin by using and maintaining this qualification infrastructure.

The active design is documented in `docs/qualification/PULMONARY_CAPILLARY_QUALIFICATION_PROTOCOL.md`.



| Program                                                | Principal work                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Intended outcome                                                                                                                                                                                                                 |
|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Pulmonary and capillary qualification</b>           | <b>PCQ-1.1 and PCQ-1.2 completed locally:</b> two candidate hashes, four tracks, six primary endpoints, data-role separation, uncertainty/identifiability, controls, and a thirteen-source candidate register are machine-checked. Arizona iCPET remains the primary hemodynamic request; D'Souza 2025 is prioritized for joint functional volume and flow; Lassen 2023 requires a whole-pulmonary transit observation model; Bailey 2019 is the next five-lobe inquiry; PVDOMICS is not eligible as the initial healthy invasive source under its published protocol. PCQ-1.3 must lock observation models and tolerances before participant records. | Present result: auditable design and source selection, not a validation pass. Intended exit: bounded pulmonary and capillary claims evaluated without refitting, with uncertainty and all partial or negative findings retained. |
| <b>Biological cell-model qualification</b>             | Select a named ligand, receptor, cell type, and reusable public model/data set; reproduce the published behavior before mapping it to MEHLISSA.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | At least one biologically evidence-backed M5 variant while synthetic models remain explicit software fixtures.                                                                                                                   |
| <b>Dynamic capillary-tissue-cell coupling</b>          | Replace the prescribed homogeneous signal snapshot with time-dependent transport, binding or consumption, feedback, and conservation checks.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Cell exposure generated by a qualified coupled transport model rather than an imposed input alone.                                                                                                                               |
| <b>Externally validated medical reference scenario</b> | Freeze one measurable complete protocol and compare its injection-to-measurement outputs with independent physiological or experimental observations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | The first reproducible end-to-end validation report with a bounded claim and explicit limitations.                                                                                                                               |
| <b>Additional medical scenarios</b>                    | Continuous monitoring, liquid biopsy, endocrine/adrenal venous sampling, CAR-T, and ultimately metastasis prevention.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Evidence that the architecture generalizes beyond fingerprinting.                                                                                                                                                                |
| <b>Additional organs</b>                               | Begin with a kidney surrogate and progress toward regional filtration, clearance, and organ-specific capillary interfaces.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Evidence that organ factories and coupling contracts are not lung-specific.                                                                                                                                                      |
| <b>Personalization and digital twin</b>                | Patient manifest, imaging and SimVascular/CFD import, parameter estimation, identifiability, longitudinal updates, and governance.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Research-twin maturity from geometry to physiology and biochemistry.                                                                                                                                                             |
| <b>Scaling and HPC</b>                                 | Bound output, profile CPU/memory/I/O, use population and field representations, and parallelize safe work.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Large ensembles and sensitivity studies without sacrificing validated behavior.                                                                                                                                                  |
| <b>Advanced analysis and visualization</b>             | Extend the implemented dashboards and exports with qualified ensemble statistics, uncertainty propagation, spatiotemporal views, and larger-study comparison.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Evidence-backed analyses for study designs and scales beyond the bounded Workbench 1.0 demonstrators.                                                                                                                            |

The roadmap priority tiers have the following meaning in the current state:

- **P0, indispensable foundation, is substantially complete:** kernel, reproducibility, CI, units, schemas, and the BVS baseline.
- **P1, dissertation core, is substantially complete at demonstrator level:** body, lung, capillary, cell, Nano-IoT, and fingerprinting vertical slice.
- **P2, platform expansion, is now the main scientific expansion space:** new qualification evidence, scenarios, multiple organs, external models, and advanced scientific visualization beyond Workbench 1.0.
- **P3, long-term vision, remains open:** metastasis prevention, patient-specific models, a dynamic digital twin, HPC ensembles, and eventual clinical validation where appropriate.

## 8. Integrated research-use capabilities and delivery record

UX means user experience and identifies the implementation history of MEHLISSA's research-use layer. The resulting commands, reports, campaign tools, Python interfaces, notebooks, and Workbench 1.0 are supported parts of the platform, not a separate future program. They make the scientific capabilities operable, reproducible, and auditable, but do not by themselves increase physiological or clinical validity.

| Package                                                             | Plain-language objective                                                                                                                                                               | Status                                                                                                                                          |
|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>UX-1 - One-command M7 scenario execution</b>                     | Validate and run the complete fingerprinting demonstrator and summarize its result through the normal application.                                                                     | Passed; all 280 local Windows/MSVC tests and all supported GitHub CI jobs pass                                                                  |
| <b>UX-2 - Model and example discovery</b>                           | List and describe available artifacts, parameters, evidence, and limitations.                                                                                                          | Passed; five model families, ten examples, all 281 local Windows/MSVC tests, and all supported GitHub CI jobs pass                              |
| <b>UX-3 - Human-readable and HTML result reporting</b>              | Provide concise terminal, tabular, and shareable HTML views over the complete machine-readable result.                                                                                 | Passed; six-file bundle, all 284 current tests, and all supported GitHub CI jobs pass                                                           |
| <b>UX-4 - Derived experiments and campaigns</b>                     | Create controlled variants, replicates, parameter sweeps, paired comparisons, and aggregate analyses.                                                                                  | Passed; six-run reference campaign, all 284 current tests, and all supported GitHub CI jobs pass                                                |
| <b>UX-5 - Python API and notebooks</b>                              | Support common scientific analysis and plotting workflows without replacing the C++ implementation authority.                                                                          | Passed; process API, readers, analysis, optional plots, two notebooks, all 284 current tests, and all supported GitHub CI jobs pass             |
| <b>UX-6.1 - Workbench product and technical foundation</b>          | Define users and workflows, select the interface architecture, establish safety/accessibility baselines, and prove read-only graphical discovery without duplicating scientific logic. | Passed and included in Workbench 1.0; five model families and ten examples load through the accepted process API                                |
| <b>UX-6.2 - Guided scenario workspace</b>                           | Open the complete FP9/lung starter, edit schema-derived scalar fields with their units/evidence/limits, retain the complete source, and validate/save/reopen without overwriting.      | Passed and included in Workbench 1.0; a derivative round-trips with all 13 artifacts/evidence retained and unsafe writes fail closed            |
| <b>UX-6.3 - Validation and corrective feedback</b>                  | Explain structural, semantic, and cross-file problems before save/run and make the decision shareable.                                                                                 | Passed and included in Workbench 1.0; accepted-CLI parity, stable located diagnostics, candidate identity, and closed invalid save/run gates    |
| <b>UX-6.4 - Run and campaign control</b>                            | Confirm, start, monitor, cancel, and inspect an exact validated scenario or the curated six-run campaign without bypassing accepted APIs.                                              | Passed and included in Workbench 1.0; bounded evidence, exact inputs/seeds, lifecycle, real cancellation, and protected artifacts               |
| <b>UX-6.5 - Result dashboard and comparison</b>                     | Explain completed outcomes, cases, stages, campaign groups, and paired differences while retaining authoritative artifacts and excluding non-results.                                  | Passed and included in Workbench 1.0; accepted-reader parity, guarded comparison/reporting, and explicit zero-observation failure policy        |
| <b>UX-6.6 - Provenance, evidence, and interpretation boundaries</b> | Keep software/build identity, seeds, hashes, models, sources, licences, maturity, limitations, and the non-clinical scope attached to every displayed result.                          | Passed and included in Workbench 1.0; round-trip provenance, integrity/evidence states, and exact JSON audit export                             |
| <b>UX-6.7 - Sensitivity, uncertainty, visualization, and export</b> | Explain replicate variation, parameter effects, and same-seed differences without overstating the small curated campaign, then export the exact analysis.                              | Passed and included in Workbench 1.0; accepted-reader observations, descriptive views, exact-value table, and source-bound JSON/CSV/SVG exports |

| Package                                                                     | Plain-language objective                                                                                       | Status                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>UX-6.8 - Usability, accessibility, packaging, and release acceptance</b> | Package and test the workbench as a dependable entry point for novice and expert researchers and contributors. | Passed; version 1.0 wheel/command, isolated install, example workspace, semantic accessibility and responsive recovery gates, 286-test regression, synchronized documentation, and green supported CI for the exact published commit |

The Workbench is the recommended integrated entry point:

```
mehlissa-workbench --repository-root . --check
mehlissa-workbench --repository-root .
```

The equivalent composable command-line route remains available for automation, method inspection, and advanced workflows:

```
mehlissa scenario list
mehlissa scenario validate --file examples/scenarios/fp9-lung-level-a-v1.json
mehlissa scenario run --file examples/scenarios/fp9-lung-level-a-v1.json --output
results/fp9-reference
mehlissa result summarize --file results/fp9-reference/<printed-run-directory>/result.json
mehlissa result report --file results/fp9-reference/<printed-run-directory>/result.json --output
reports/fp9-reference
mehlissa campaign run --file examples/campaigns/fp9-collector-count-v1.json --output
results/fp9-campaign
```

UX-1 and UX-2 established the operational entry layer: one-command execution, actionable errors, unique result directories, model and example discovery, and fail-safe starter copying. UX-3 through UX-5 added the reproducible research workflow: non-overwriting reports, retained campaign variants, deterministic replicates and paired comparisons, version-guarded Python readers, optional plots, and two licensed notebooks. Inputs, hashes, seeds, evidence, limitations, and the non-clinical boundary remain attached throughout.

UX-6 delivered the graphical layer as MEHLISSA Research Workbench 1.0.0. Its eight increments cover product and safety foundations, schema-derived scenario editing, accepted-command validation, confirmed execution and cancellation, reader-backed dashboards and comparisons, provenance and evidence audit, descriptive campaign analysis and exact export, accessibility, packaging, and release acceptance. The local browser host delegates scientific authority to the matching C++ simulator, repository artifacts, and accepted Python process API. It binds only to the local computer, serves no remote assets, records no telemetry, and cannot convert failed or incomplete jobs into observations.

The complete published Workbench release passes 286 local Windows/MSVC tests and GitHub Windows/MSVC, Linux/GCC, and Linux/Clang CI, including formatting, clang-tidy, AddressSanitizer, and UndefinedBehaviorSanitizer. Detailed increment contracts, acceptance evidence, and interpretation limits remain in docs/ux and the User Guide.

The platform now also provides an accepted native macOS/Apple Clang source build and the same Workbench workflow. Dedicated CMake presets and the pinned macOS 15 ARM64 job passed the complete suite in GitHub CI run 33956456353. This support does not create an application bundle, installer, signed or notarized binary, or downloadable executable. Intel Macs remain an unqualified source-compatibility expectation rather than an accepted CI claim.

## 9. Opportunities for collaborators

| Contributor profile                          | High-value starting point                                                                                                     |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>C++ and simulation engineering</b>        | General scenario runner, multirate orchestration, model factories, profiling, and regression infrastructure.                  |
| <b>Hemodynamics and pulmonary physiology</b> | Review pulmonary cards, qualify cohorts, refine state dependence, and develop anatomical or pulsatile variants.               |
| <b>Renal physiology</b>                      | Define and validate a kidney model family using the existing organ and capillary contracts.                                   |
| <b>Molecular communication</b>               | Calibrate local channels, implement assembly physics, and connect an external network simulator through the existing adapter. |

| Contributor profile                       | High-value starting point                                                                                                                                                       |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cell biology and pharmacology             | Replace synthetic receptor, signaling, apoptosis, and delivery assumptions with evidence-qualified models.                                                                      |
| Statistics and uncertainty quantification | Design independent validation, global sensitivity, uncertainty propagation, identifiability, and ensemble analysis.                                                             |
| Research UX and visualization             | Evaluate Workbench 1.0 with diverse researchers and assistive technologies, then turn observed friction into bounded follow-up packages without weakening scientific authority. |
| Research data and governance              | Model-card provenance, licensing, FAIR data, patient-data controls, and digital-twin governance.                                                                                |

A new module is reviewable when its public contract and units are explicit; misuse fails predictably; time, identity, ownership, and conservation invariants pass; randomness is reproducible; output is bounded; schemas and examples agree; software verification is separated from calibration and validation; evidence and limitations are documented; and all supported CI paths pass.

## 10. Key project references

| Document                                                                                            | Purpose                                                                                                                                                              |
|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| docs/USER_GUIDE.md                                                                                  | Non-technical introduction, experiment families, installation, commands, model workflows, interpretation, and troubleshooting.                                       |
| docs/architecture/SOFTWARE_ARCHITECTURE.md                                                          | System structure, APIs, extension workflow, kidney outline, external simulator integration, and personalization.                                                     |
| docs/ROADMAP.md                                                                                     | Guiding principles, delivery history, scientific qualification sequence, medical scenarios, additional organs, personalization, scaling, and cross-cutting programs. |
| docs/qualification/PULMONARY_CAPILLARY_QUALIFICATION_PROTOCOL.md and PCQ1_EVIDENCE_SOURCE_SCREEN.md | Active PCQ-1 claim, frozen candidates, endpoints, ranked sources, access/licence boundary, controls, and pre-outcome sequence.                                       |
| docs/PROJECT_STATE.json                                                                             | Machine-readable shared release facts, Roadmap section mappings, selected traceability expectations, and ordered qualification packages.                             |
| docs/requirements/SYSTEM_REQUIREMENTS.md                                                            | Numbered requirements derived from the dissertation and project decisions.                                                                                           |
| docs/requirements/TRACEABILITY_MATRIX.md                                                            | Separate implementation and evidence status, achieved verification, and remaining gap for every tracked requirement.                                                 |
| docs/m7/M7_GATE_REVIEW.md                                                                           | Formal decision and scientific limitations of the holistic fingerprinting demonstrator.                                                                              |
| docs/publication/EVIDENCE_AND_VALIDITY_BASELINE.md                                                  | Six-family machine-readable evidence and validity baseline, source-role audit, and claim boundaries.                                                                 |
| docs/publication/PAPER1_TECHNICAL_EXPERIMENT_PROTOCOL_V2.md                                         | Locked Paper 1 technical questions, conditions, controls, metrics, failure rules, and archive contract.                                                              |
| docs/publication/PAPER1_TECHNICAL_MEASUREMENTS.md                                                   | Complete body-observation, M7 resource/replay, and access-parity results with explicit non-claims.                                                                   |
| publication/paper1/release-candidates/paper1-platform-methods-rc1-20260903/HANDOFF.md               | Exact source and artifact identity, reproduction route, claim limits, and reviewer checklist.                                                                        |
| docs/ux/README.md                                                                                   | Index of all eight Workbench increments and their detailed product, validation, execution, reporting, audit, analysis, accessibility, and release contracts.         |

- **Repository:** <https://github.com/RegineWendt/MEHLISSA>
- **Verified Workbench 1.0 CI run:** <https://github.com/RegineWendt/MEHLISSA/actions/runs/33745263319>